

Lab program-16:

16. Develop a C program for implementing random access file for processing the employee details.

Code:

```
#include <stdio.h>
#include <stdlib.h>

struct Employee
{
    int id;
    char name[50];
    float salary;
};

int main()
{
    FILE *fp;
    struct Employee emp;

    int choice;
    int id;
    long position;

    fp = fopen("employee.dat", "rb+");

    if (fp == NULL)
    {
        fp = fopen("employee.dat", "wb+");

        if (fp == NULL)
        {
            printf("Error opening file.\n");
            return 1;
        }
    }

    while (1)
    {
        printf("\n1. Add Employee");
        printf("\n2. Display Employee");
        printf("\n3. Exit");
        printf("\nEnter your choice: ");

        scanf("%d", &choice);
```

```

switch (choice)
{
    case 1:
        printf("\nEnter Employee ID: ");
        scanf("%d", &emp.id);

        printf("Enter Employee Name: ");
        scanf("%s", emp.name);

        printf("Enter Employee Salary: ");
        scanf("%f", &emp.salary);

        position = (emp.id - 1) * sizeof(struct Employee);

        fseek(fp, position, SEEK_SET);

        fwrite(&emp, sizeof(struct Employee), 1, fp);

        printf("Employee record saved successfully.\n");
        break;

    case 2:
        printf("\nEnter Employee ID to search: ");
        scanf("%d", &id);

        position = (id - 1) * sizeof(struct Employee);

        fseek(fp, position, SEEK_SET);

        if (fread(&emp, sizeof(struct Employee), 1, fp) == 1 &&
            emp.id == id)
        {
            printf("\nEmployee Details\n");
            printf("ID      : %d\n", emp.id);
            printf("Name   : %s\n", emp.name);
            printf("Salary : %.2f\n", emp.salary);
        }
        else
        {
            printf("Employee record not found.\n");
        }

        break;

    case 3:
        fclose(fp);
        return 0;
}

```

```
        default:  
            printf("Invalid choice.\n");  
    }  
}  
}
```

OUTPUT:

● PS C:\Users\SMILE\Documents\OS\Lab programs> ./lab_16.exe

```
1. Add Employee  
2. Display Employee  
3. Exit  
Enter your choice: 1
```

```
Enter Employee ID: 123  
Enter Employee Name: eee  
Enter Employee Salary: 3000  
Employee record saved successfully.
```

```
1. Add Employee  
2. Display Employee  
3. Exit  
Enter your choice: 2
```

```
Enter Employee ID to search: 123
```

```
Employee Details  
ID      : 123  
Name    : eee  
Salary  : 3000.00
```

```
1. Add Employee  
2. Display Employee  
3. Exit  
Enter your choice: 3
```